



Robust adhesive Janus hydrogel reinforced by topological entanglement for abdominal wall defect therapy

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ABSTRACT

Adhesive hydrogels hold promise for seamless closure of abdominal wall defects. However, most of them suffer from an adhesion-cohesion conflict, rendering them susceptible to swelling and fracture within the biofluid-rich and mechanically dynamic abdominal environment. This study developed a topological entanglement-reinforced Janus BGMA/BPEG hydrogel for abdominal wall defect treatment. The hydrogel is formed by physically immobilizing gelatin methacryloyl or polyethylene glycol diacrylate polymer chains within a bacterial cellulose nanofibril framework via hydrogen-bonding remodeling, followed by *in situ* photo-triggered chemical crosslinking. This topological entanglement structure prevents efflux of polymer chains from the defect site while maintaining sufficient chain mobility prior to crosslinking. This facilitates strong interfacial binding between the BGMA side and abdominal tissues upon photo-crosslinking. The BGMA side also promotes fibroblast spreading and proliferation to accelerate wound healing. The opposing BPEG side, with higher hydrophilicity, inhibits visceral adhesions. Additionally, the topological entanglement structure constructs a pathway for efficient load transfer and constrains chain relaxation, significantly enhancing the toughness and water resistance. This reinforcement enables the hydrogel to maintain structural integrity and protect interfacial adhesion within the harsh abdominal environment. Overall, this topological entanglement-reinforced Janus hydrogel demonstrates both enhanced adhesion and cohesion, highlighting its potential as a preclinical platform for abdominal defect treatment.

1. Introduction

Abdominal wall defects, caused by trauma, tumor resection, or severe infections, present significant clinical challenges [1,2]. The loss of abdominal wall tissue compromises the natural barrier of the abdominal cavity, critically impairing its mechanical support and intrinsic lubrication functions [3]. This disruption exposes the underlying viscera and triggers a localized inflammatory response. This frequently results in intestinal adhesions, leading to substantial chronic abdominal pain and secondary bowel obstruction. As the condition progresses without adequate intervention, patients face a cascade of debilitating and potentially life-threatening complications [4–6]. The current gold standard for clinical treatment is tension-free surgery, which involves a biomaterial patch to separate the damaged tissue from adjacent organs

to prevent fibrin bridge formation [7,8]. However, commercially available patches typically require suturing for fixation, leaving gaps between the sutured patch and the abdominal wall tissue that increase the risk of infection and visceral adhesions [9,10]. Furthermore, these patches are primarily composed of inert materials (e.g., polypropylene, polyester), failing to actively promote wound healing [11–13]. Consequently, there is an urgent need to develop biomaterials capable of seamlessly closing abdominal wall defects while simultaneously integrating anti-adhesion and pro-healing functionalities.

Biopolymer-based adhesive hydrogels have garnered increasing attention for seamless abdominal wall defect closure, owing to their strong interfacial binding ability, excellent biocompatibility, and tunable biological activity [14,15]. Current adhesive hydrogels for wound closure primarily fall into two categories. The first category

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consists of pre-formed hydrogel patches [16,17], which typically require low crosslinking densities to maintain sufficient polymer chain mobility for interfacial binding. However, this reduced density compromises swelling resistance, causing hydrogel expansion in the biofluid-rich abdominal cavity and ultimately leading to interfacial failure and detachment. Although increasing the crosslinking density mitigates hydrogel swelling [18], it significantly restricts polymer chain mobility and undermines adhesion strength. The second category involves *in situ*-formed hydrogels, generated by applying precursor solutions or cryo-hydrogel powders directly to the defect site followed by *in situ* crosslinking [19,20]. This approach achieves stronger adhesion strength due to favorable polymer chain mobility prior to chemical crosslinking [21]. Nevertheless, precursor solutions or powders fail to be retained at the defect site due to the lack of mechanical support, particularly in large abdominal wall defects. Additionally, most adhesive hydrogels exhibit inferior mechanical properties due to the inherent conflict between cohesion and adhesion, raising concerns regarding structural fracture under abdominal pressure or deformation [22,23]. Thus, developing adhesive hydrogels that can seal large wounds and simultaneously provide robust mechanical properties and anti-swelling capacity is essential for the treatment of abdominal wall defects.

Bacterial cellulose (BC), a biopolymer with an inherent nanofibrillar interwoven network secreted from *Acetobacter xylinum*, is a promising wound dressing material [24,25]. It possesses strong yet flexible mechanical performance, attributed to strong intra- and inter-molecular hydrogen bonds in the crystalline region [26,27]. However, excessive internal hydrogen bonding of BC limits its binding affinity to external substrates, resulting in negligible adhesion strength to biological tissues. Current strategies for fabricating BC-based adhesive hydrogels involve modulating hydrogen bonds via ionic liquids or dissolving BC before incorporating into an adhesive matrix [28]. Nevertheless, ionic liquids readily exchange with biofluids post-implantation, compromising both adhesiveness and biosafety, while BC dissolution severely degrades its mechanical performance. Our recent research reports an adhesive yet robust BC-based hydrogel featuring a topological entanglement structure [29], defined as the purely geometric entangling between polymer chains and templates resulting from their spatial interpenetration, interweaving, and looping [30]. This unique architecture is achieved by infiltrating oxygen-rich biopolymers into the BC framework, where these flexible polymer chains geometrically wrap around the rigid BC nanofibrils and form extensive hydrogen bonds, generating robust entanglements that span from the molecular to the nanoscale. Crucially, this topological structure preserves the chain flexibility required for interfacial adhesion while simultaneously enhancing mechanical properties through cross-scale load transfer. Hence, immobilizing adhesive biopolymers within the BC framework via topological entanglement, followed by *in situ* crosslinking, holds significant promise for fabricating robust adhesive hydrogels for abdominal wall defect repair.

In this study, a hydrogen-bonding remodeling strategy was proposed to drive the topological entanglement of gelatin methacryloyl (GelMA) or polyethylene glycol diacrylate (PEGDA) polymer chains within a BC framework, yielding BC-immobilized GelMA (BGMA) and BC-immobilized PEGDA (BPEG) pre-gel intermediates. Upon light irradiation, these pre-gels underwent *in situ* conversion at wound sites into an asymmetrically adhesive, anti-swelling, and robust Janus BGMA/BPEG hydrogel. In a full-thickness abdominal wall defect rat model, the Janus hydrogel strongly adhered to the defect site via its *in situ*-formed BGMA side, effectively sealing the wounds and accelerating healing. In contrast, the highly hydrophilic BPEG side repelled cell attachment, thereby preventing visceral adhesions. Furthermore, the topological entanglement structure allowed the hydrogel to preserve interfacial adhesion, maintain structural integrity, and restrict swelling within the biofluid-rich and mechanically dynamic abdominal environment throughout the 14-day treatment period.

2. Results

2.1. Design strategy

The fabrication of the robust Janus BGMA/BPEG hydrogel for abdominal wall defect treatment involves two steps. First, a hydrogen-bonding remodeling strategy is employed to prepare soft BGMA and BPEG pre-gel intermediates exhibiting topological entanglement characteristics (Fig. 1A). These pre-gels are formed by incorporating *Acetobacter xylinum*-secreted BC into photocrosslinkable GelMA or PEGDA precursors containing the photo-initiator lithium phenyl-2,4,6-trimethylbenzoylphosphine (LAP). During this process, GelMA or PEGDA chains with rich oxygen-containing groups serve as hydrogen bond donors/acceptors, which competitively disrupt intrinsic hydrogen bonds of BC molecules in the crystalline region and establishes new hydrogen bonds. This rearrangement of the hydrogen bond network drives the formation of topological entanglement between polymer chains and BC nanofibrils in the pre-gels. The resulting topological entanglement structure physically immobilizes GelMA and PEGDA chains within the BC framework, preventing their free diffusion at the abdominal defect site while maintaining sufficient mobility for interfacial binding in the subsequent process.

Second, a photo-triggered *in situ* chemical crosslinking process transforms the stacked BGMA and BPEG pre-gels into a robust Janus BGMA/BPEG hydrogel at the abdominal defect site (Fig. 1B). Prior to crosslinking, the relatively mobile GelMA chains in the BGMA pre-gel and the PEGDA chains in the BPEG pre-gel interpenetrate at the interface. Upon photo irradiation (395 nm), GelMA and/or PEGDA chains form covalent C—C bonds, permanently fixing the topological entanglement within the hydrogel matrix and establishing strong interfacial bonding between the two layers. The mobile GelMA chains in the pre-gel also enable the crosslinked BGMA side to establish conformal contact and form multiple interfacial bonds with abdominal tissue, achieving seamless wound closure (Fig. 1C). The topological entanglement structure significantly enhances the cohesion of the Janus hydrogel. It not only allows efficient load transfer and energy dissipation, thereby enhancing toughness (Fig. 1D), but also topologically constrains the relaxation of GelMA and PEGDA chains, leading to improved swelling resistance (Fig. 1E). The enhanced cohesion facilitates the Janus hydrogel to maintain structural integrity and protect the interfacial adhesion within the bio-fluid rich and mechanically dynamic abdominal environment. In addition, the Janus BGMA/BPEG hydrogel exhibits asymmetric biological activities (Fig. 1F). The BGMA side promotes fibroblast spreading and proliferation via RGD sequences in GelMA chains, accelerating wound healing. However, the BPEG side presents a highly hydrophilic surface that strongly inhibits the adhesion of proteins and cells. Consequently, a mechanically robust Janus hydrogel with a tissue/cell adhesive BGMA side and an anti-adhesive BPEG side is constructed.

2.2. Hydrogen-bonding remodeling-driven topological entanglement in pre-gels

The BGMA and BPEG pre-gels were formed by incorporating the GelMA or PEGDA chains into the BC nanofibril template (Fig. 2A). As demonstrated in Fig. 2B, the BGMA and BPEG pre-gels could be lifted with tweezers and maintained their structural integrity, suggesting that the highly mobile GelMA and PEGDA chains were physically immobilized within the BC nanofibril framework. Dynamic rheological analysis (Fig. 2C) performed at 1% strain and 1 Hz frequency revealed solid-like behavior of the BGMA and BPEG pre-gels, evidenced by their relatively higher storage modulus (G') than loss modulus (G''). This contrasted sharply with the liquid-like behavior of the GelMA and PEGDA precursors. Notably, the G' values of the BGMA and BPEG pre-gels also exceeded that of pure BC, suggesting an enhanced physical crosslinking density between GelMA or PEGDA chains and the BC nanofibrils within

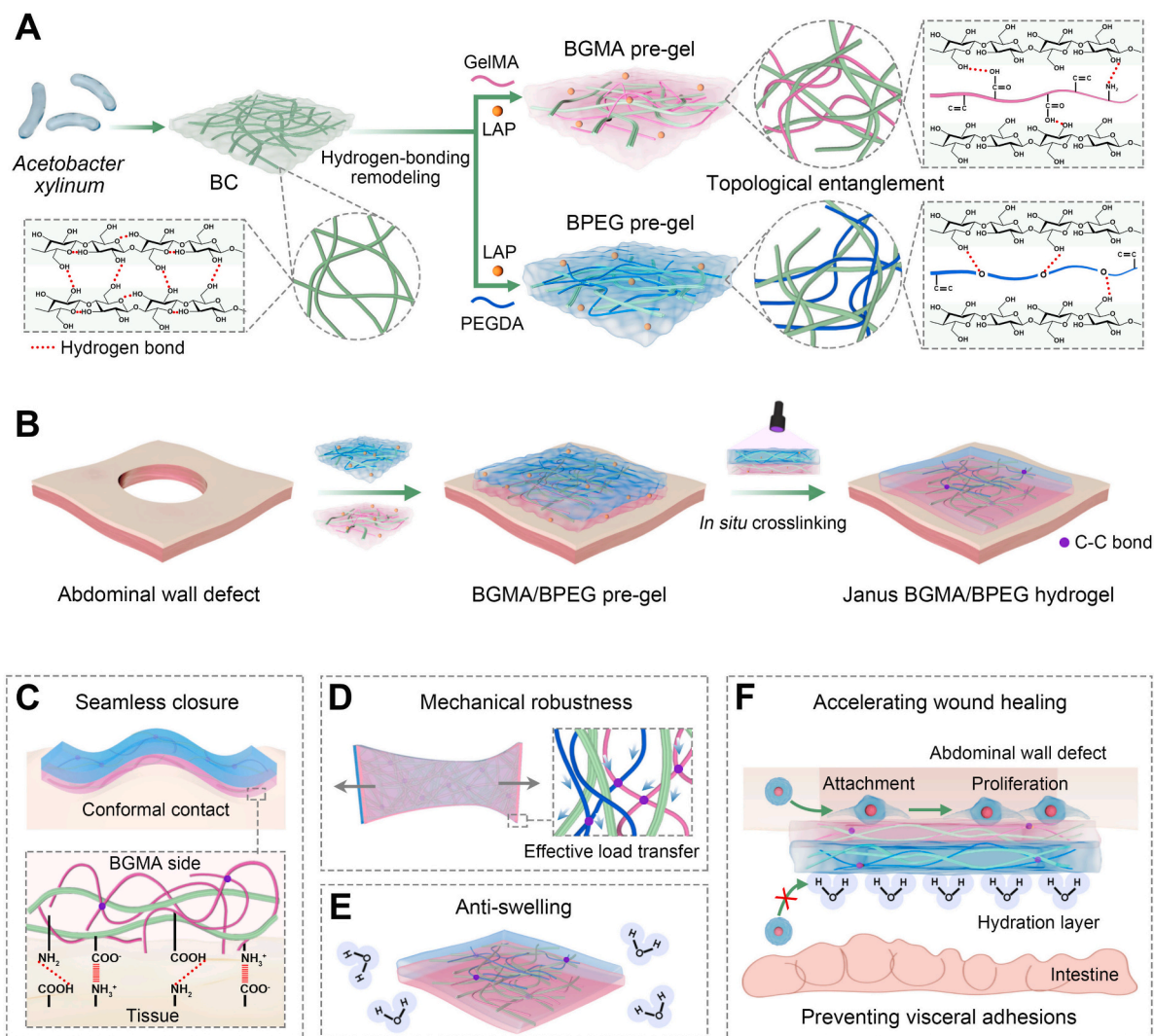


Fig. 1. Schematic diagram of the Janus BGMA/BPEG hydrogel with topological entanglement structure for abdominal defect treatment. (A) Synthesis of the BGMA and BPEG pre-gels via hydrogen-bonding remodeling. (B) Illustration of the *in situ* formed Janus BGMA/BPEG hydrogel converted from pre-gels. (C) The hydrogel forms conformational contact and multiple bonds with abdominal tissues for seamless wound closure. (D) The topological entanglement structure effectively transfers load to endow the hydrogel with mechanical robustness. (E) Anti-swelling behavior of the hydrogel. (F) The hydrogel accelerates wound healing by promoting fibroblast attachment and proliferation, while preventing visceral adhesions.

the pre-gels.

X-ray diffraction (XRD) analysis (Fig. 2D) showed that the BGMA and BPEG pre-gels displayed the same crystalline structure as pure BC, indicated by characteristic peaks at 14.4° , 16.6° and 22.6° corresponding to the (1–10), (110), and (200) planes of cellulose Ia [31,32], respectively. However, the crystallinity of BC within the BGMA and BPEG pre-gels decreased significantly from 74% (pure BC) to 25% or 36%, respectively (Fig. 2E, Fig. S1). This reduction indicates a partial disruption of intra- and inter-molecular hydrogen bonding in the BC upon the incorporation of GelMA or PEGDA. Differential scanning calorimetry (DSC) analysis indicated a lower melting temperature (1.2°C) for the BGMA pre-gel compared with the GelMA precursor (1.8°C) and BC (4.1°C) (Fig. 2F). A similar decrease in melting temperature was observed for the BPEG pre-gel (0.4°C), which was also lower than that of the PEGDA precursor (0.9°C) and BC (4.1°C). The decreased melting temperature of the pre-gels is primarily due to the rearrangement of the hydrogen-bonding network following the incorporation of the GelMA or PEGDA chains. Direct evidence of these hydrogen-bonding interactions was provided by Fourier transform infrared spectroscopy (FTIR). The spectrum of the BGMA pre-gel is similar to that of GelMA, exhibiting characteristic peaks at 3286, 2935,

1630, and 1526 cm^{-1} , which correspond to O–H/N–H stretching, C–H stretching, C=O stretching, and N–H bending, respectively. This similarity is due to the significantly higher proportion of GelMA compared to BC within the BGMA pre-gel. Furthermore, in the magnified region ranging from 3500 to 3000 cm^{-1} , the BGMA pre-gel demonstrates a markedly broader absorption peak centered around 3286 cm^{-1} when compared to pristine BC and the GelMA (Fig. S2). This result confirms the presence of richer and more extensive hydrogen-bonding interactions occurring within the BGMA pre-gel network.

To further demonstrate the critical role of hydrogen bonding in performance enhancement, urea, a typical hydrogen-bond disruptor, was incorporated into the pre-gels and subsequently evaluated via rheological analysis (Fig. S3). The pristine BGMA pre-gel exhibited a higher G' than the urea-incorporated BGMA pre-gel. Similarly, the G' of the pristine BPEG pre-gel was higher than that of the urea-incorporated BPEG pre-gel. These results indicate that urea disrupted the hydrogen bonding between the GelMA/PEGDA chains and the BC nanofibrils, leading to decreased mechanical performance. Based on the aforementioned experimental characterizations, we hypothesize the synthesis mechanism as follows. The GelMA and PEGDA chains competitively bind to BC nanofibrils via hydrogen bonds, driving the formation of a

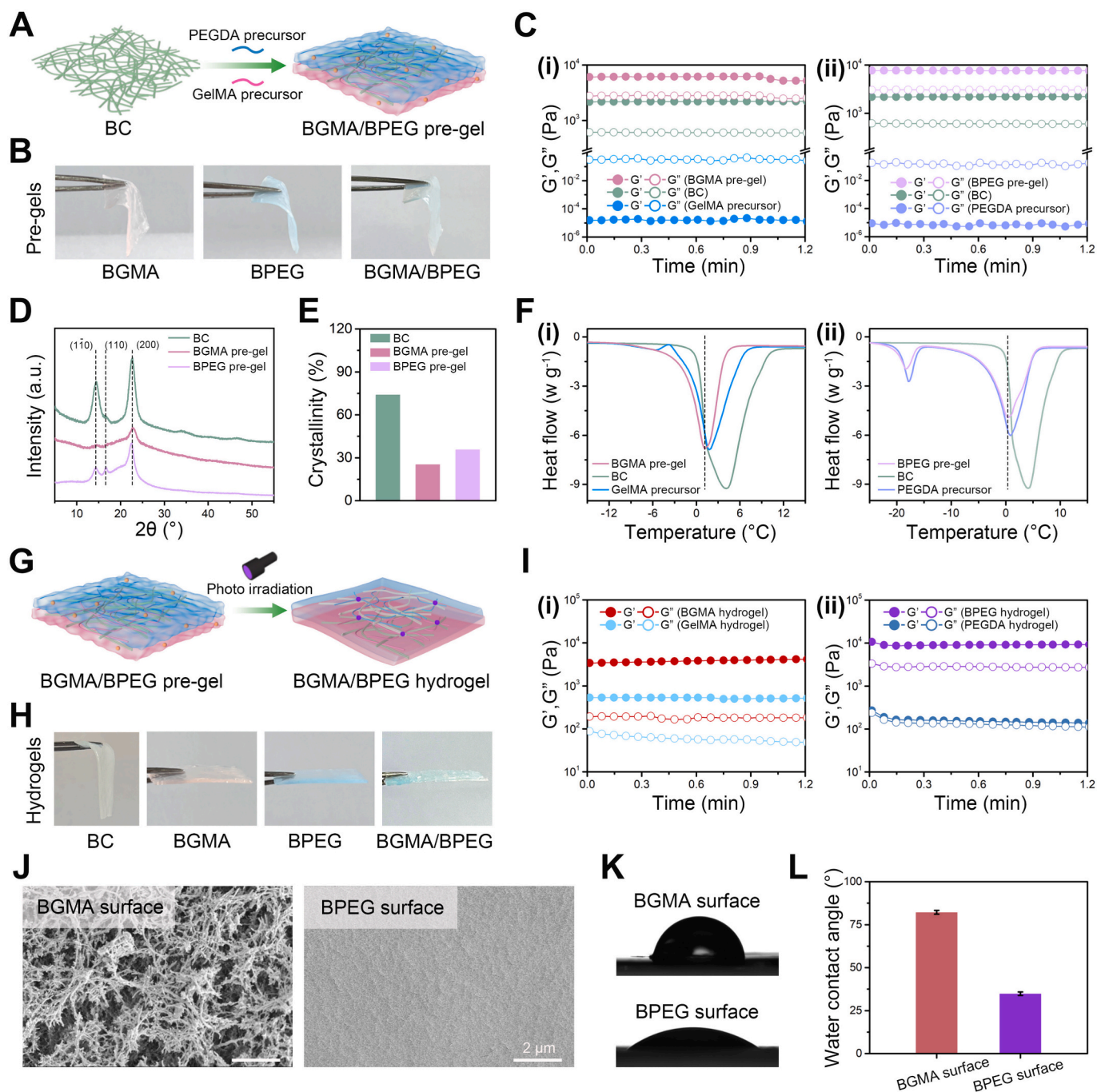


Fig. 2. Characterization of the pre-gels and Janus BGMA/BPEG hydrogel with topological entanglement structure. (A) Illustration of the formation of BGMA/BPEG pre-gels. (B) Digital photos of soft pre-gels lifted with tweezers, without efflux of polymer precursors. (C) Dynamic rheological results of the (i) BGMA pre-gel and (ii) BPEG pre-gel. (D) XRD spectra and (E) crystallinity of the pre-gels. (F) DSC results revealed the change in melting temperature of the (i) BGMA pre-gel and (ii) BPEG pre-gel. (G) Illustration of the formation of the BGMA/BPEG hydrogel converted from pre-gels upon photo irradiation. (H) Digital photos of the hydrogel with enhanced stiffness. (I) Dynamic rheological results of the (i) BGMA hydrogel and (ii) BPEG hydrogel. (J) SEM images of the opposing side of the Janus BGMA/BPEG hydrogel. (K) Digital photos of water droplets on the BGMA and BPEG hydrogel surfaces. (L) Water contact angles of the Janus hydrogel.

topological entanglement structure and thereby enhancing the physical crosslinking density and mechanical modulus.

2.3. Formation of the Janus hydrogels with topological entanglement characteristics

The Janus BGMA/BPEG hydrogel, converted from the pre-gels upon photo irradiation, retains the topological entanglement features (Fig. 2G). Fig. 2H demonstrated increased stiffness in BGMA, BPEG, and

Janus BGMA/BPEG hydrogels compared to their respective pre-gels and pristine BC, revealing a transition from soft pre-gels to robust hydrogels following chemical crosslinking. Dynamic rheological analysis further showed that the G' of BGMA and BPEG hydrogels exceeded those of GelMA and PEGDA hydrogels, respectively (Fig. 2I). This indicates that the topological entanglement-induced enhancement observed in pre-gels were preserved in the crosslinked hydrogels.

The Janus BGMA/BPEG hydrogel possesses asymmetric morphology and hydrophilicity on its opposing sides. Scanning electron microscopy

(SEM) images (Fig. 2J) indicated that the BGMA side displayed a loose and porous structure with interwoven BC nanofibrils. By contrast, the BPEG side presented a markedly denser morphology. Water contact angle measurements confirmed this asymmetry (Fig. 2K, L), with the BPEG side ($34.8 \pm 1^\circ$) exhibiting higher hydrophilicity than the BGMA side ($82.3 \pm 1^\circ$), owing to the abundant ether bonds in the PEGDA backbone [33]. These results reveal the relatively loose, hydrophobic BGMA side and the relatively dense, hydrophilic BPEG side of the Janus hydrogel.

The interfacial binding between the BGMA and BPEG layers of the Janus hydrogel was visualized using a fluorescent microscope. To facilitate tracking, rhodamine and fluorescein isothiocyanate (FITC) were separately incorporated into the BGMA and BPEG pre-gels prior to crosslinking. As shown in Fig. S4A, rhodamine and FITC signals were

distributed on the upper and lower regions, which corresponded to the BGMA and BPEG layers, respectively. An overlapping of these signals was observed at the intermediate region, suggesting the tight binding between the BGMA and BPEG layers of the Janus hydrogel. SEM confirmed continuous connection between the two layers at the cross-section of the Janus hydrogel (Fig. S4B). Furthermore, a lap-shear test quantified the interfacial binding strength at 140 ± 3 kPa (Fig. S4C). The interfacial binding of the Janus hydrogel was further evaluated following 14 days of *in vivo* implantation. The Janus hydrogel maintained its structural integrity, and the two hydrogel layers remained firmly attached with no observable signs of macroscopic delamination (Fig. S5A). SEM imaging further confirmed this continuous contact at the microscopic level (Fig. S5B). Collectively, these results demonstrate that the BGMA-BPEG interface of the Janus hydrogel maintains

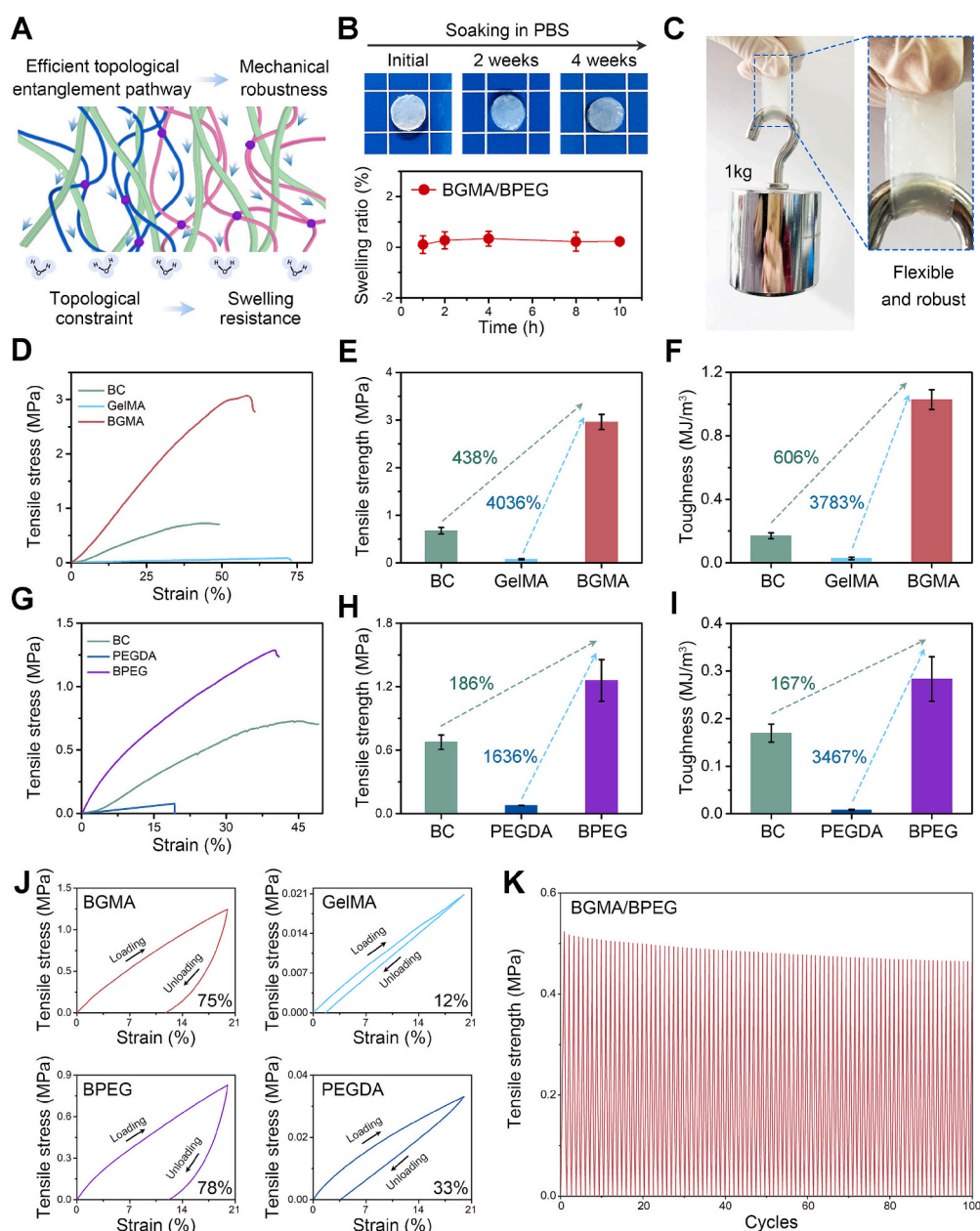


Fig. 3. Mechanical robustness and anti-swelling behaviors of the Janus BGMA/BPEG hydrogel. (A) Topological entanglement structure as efficient load transfer pathway to enhance mechanical robustness of the hydrogel, while restricting chain relaxation to improve its water resistance. (B) Digital photos of the BGMA/BPEG hydrogel soaking in PBS for 4 weeks and its swelling ratio. (C) Digital photo showing the BGMA/BPEG hydrogel withstanding a 1-kg weight. (D) Tensile curves, (E) tensile strength, and (F) toughness of the BGMA hydrogel. (G) Tensile curves, (H) tensile strength, and (I) toughness of the BPEG hydrogel. (J) Tensile loading-unloading curves of the hydrogels. (K) Cyclic tensile loading-unloading curves of the BGMA/BPEG hydrogel.

integration for at least 14 days *in vivo*.

The Janus BGMA/BPEG hydrogel exhibits *in vivo* biocompatibility and biodegradability, as demonstrated using a rat subcutaneous implantation model. Immunofluorescence staining results indicated that the skin tissues surrounding the BGMA/BPEG hydrogel showed negligible fluorescence signals of α -SMA, CD68, and iNOS, which were comparable to those of the control group (Fig. S6). This demonstrated that the hydrogel did not induce significant inflammatory or fibrotic responses, suggesting *in vivo* biocompatibility of the Janus BGMA/BPEG hydrogel. Furthermore, at 1-, 2-, 4-, and 8-weeks post-implantation, the hydrogels were also collected for biodegradability evaluation (Fig. S7). The weight of the Janus BGMA/BPEG hydrogel gradually decreased over the implantation period. The samples exhibited $78.8\% \pm 7.0\%$, $65.2\% \pm 1.5\%$, $61.5\% \pm 6.2\%$, and $57.5\% \pm 1.2\%$ mass remaining at 1-, 2-, 4- and 8-weeks post-implantation, respectively, confirming the *in vivo* biodegradability of the hydrogel.

2.4. Robust mechanical properties and anti-swelling behaviors of the Janus hydrogels

The Janus BGMA/BPEG hydrogel exhibits mechanical robustness and swelling resistance, due to the effective energy dissipation pathway and chain constraint effect by the topological entanglement structure, as well as the structural fixation caused by covalent crosslinking (Fig. 3A). The Janus BGMA/BPEG hydrogel showed minimal swelling in phosphate buffered saline (PBS), reaching an equilibrium swelling ratio of only $0.22\% \pm 0.09\%$ (Fig. 3B). Photos in Fig. 3C indicated that the BGMA/BPEG hydrogel could withstand a 1-kg weight, equivalent to 500 times its own weight. Mechanical properties of the hydrogels were quantified using a tensile test (Fig. 3D). The BGMA hydrogel exhibited a maximum tensile strength of 2.96 ± 0.16 MPa, which was 4.4-fold and 40-fold higher than those of the pure BC and GelMA hydrogels, respectively (Fig. 3E). In addition, the toughness of the BGMA hydrogel was approximately 6 and 38 times higher than that of the BC and GelMA hydrogels (Fig. 3F). Similarly, the BPEG hydrogel showed higher tensile strength and toughness, compared with the BC (1.9-fold for tensile strength, 1.7-fold for toughness) and PEGDA hydrogels (16.4-fold for tensile strength, 34.7-fold for toughness) (Fig. 3G-I). The mechanical performance of the Janus BGMA/BPEG hydrogel was intermediate between those of the BGMA and BPEG hydrogels, with a maximum tensile strength of 1.55 ± 0.13 MPa and a toughness of 0.36 ± 0.05 MJ/m³ (Fig. S8). Tensile loading-unloading curves indicated that the BGMA (75%) and BPEG hydrogel (78%) exhibited larger hysteresis rates compared with GelMA (12%) and PEGDA (33%) hydrogels, respectively, suggesting greater energy dissipation in the hydrogels with topological entanglement structures (Fig. 3J). Furthermore, the BGMA/BPEG hydrogel demonstrated good recoverability, with only 10% reduction in tensile strength after 100 loading-unloading cycles (Fig. 3K). These results demonstrate the mechanical robustness and anti-swelling properties of the Janus BGMA/BPEG hydrogel, which is conducive to withstanding abdominal deformation and maintaining the structural stability in the internal wet environment.

2.5. Asymmetric tissue adhesion performances of the Janus hydrogels

The Janus BGMA/BPEG hydrogel exhibits strong adhesion to abdominal tissues on the *in situ*-formed BGMA side while providing lubrication on the opposite BPEG side. As presented in Fig. 4A, the BGMA/BPEG hydrogel adhered tightly to abdominal tissues without detachment, even when twisted or bent after 24 h of immersion in PBS. The adhesion interface was investigated by SEM and hematoxylin and eosin (H&E) histological staining, which revealed a conformal and gapless interface between the BGMA/BPEG hydrogel and abdominal tissue (Fig. 4B).

The adhesion strength of the *in situ*-formed BGMA hydrogel to biological tissues was quantified using a lap-shear test, with a pre-prepared

BGMA hydrogel (non-*in situ*) serving as a control (Fig. 4C). As demonstrated in Fig. 4D-F, the *in situ*-formed BGMA hydrogel presented shear adhesion strengths of 14.4 ± 3.3 kPa to porcine skin and 39.0 ± 5.0 kPa to peritoneum, which were dramatically higher than those of the non-*in situ*-formed BGMA hydrogel (0.6 ± 0.1 kPa for porcine skin, 2.7 ± 0.7 kPa for peritoneum). Furthermore, the shear adhesion strength of the *in situ*-formed BGMA hydrogel increased with higher concentrations of GelMA, reaching a maximum value at 40 wt% GelMA (Fig. S9). A burst pressure test was performed to further evaluate the adhesion performance of the hydrogel (Fig. 4G). The *in situ*-formed BGMA hydrogel withstood a maximum burst pressure of 445 ± 104 mmHg, exceeding that of the non-*in situ*-formed BGMA hydrogel (8 ± 2 mmHg) and typical intra-gastrointestinal pressure (3.7–30 mmHg) [34] (Fig. 4H). Increasing GelMA concentration also enhanced burst pressure resistance of the BGMA hydrogel (Fig. S10). The robust adhesion of the BGMA side of the Janus hydrogel is attributed to the following three factors (Fig. 4I). First, GelMA chains physically immobilized within the pre-gel maintained sufficient flexibility and mobility, enabling conformal contact and establishing multiple interfacial bonds with tissues. Second, the subsequent covalent crosslinking provides structural fixation, preventing the displacement of polymer chains from tissues when the hydrogel deforms. Third, the topological entanglement between gelatin chains and BC nanofibrils facilitated efficient load transfer and energy dissipation, providing robust cohesion to protect interfacial bonds from damage. In short, the Janus BGMA/BPEG hydrogel exhibited enhanced shear adhesion strength, robust tensile strength, and a high anti-swelling coefficient. These combined properties demonstrate an overall performance that outperforms that of most previously reported adhesive hydrogel patches (Fig. S11) [35,36].

The tribological performance of the Janus BGMA/BPEG hydrogel in PBS was measured using the setup depicted in Fig. 4J. The BPEG side exhibited a lower coefficient of friction (COF) of 0.022 ± 0.002 , compared with that of the BGMA side (0.038 ± 0.013) (Fig. 4K, L). This difference is primarily attributed to the superior hydrophilicity of PEGDA, which promotes a more stable hydration layer that reduces shear friction [37].

2.6. Asymmetric cell affinity of the Janus hydrogels

The Janus BGMA/BPEG hydrogel exhibits opposite cell affinities on its two sides, attributed to asymmetric surface morphology and hydrophilicity. Initial cytocompatibility assessment involved seeding NIH-3 T3 fibroblasts on cell plates and subsequently co-culturing with the hydrogel (Fig. 5A). Live/dead staining images showed substantial spreading of viable NIH-3 T3 cells on the plates, with negligible cell death following 3-day co-culture with the BGMA/BPEG hydrogel (Fig. 5B). The MTT assay further confirmed favorable cytocompatibility of the hydrogel, showing comparable optical density (OD) values between the BGMA/BPEG hydrogel and control groups (Fig. 5C).

The cell affinity of the BGMA/BPEG hydrogel was next assessed by directly seeding NIH-3 T3 cells onto hydrogel surfaces (Fig. 5D). Confocal imaging (Fig. 5E) demonstrated significantly greater attachment of NIH-3 T3 cells on the BGMA hydrogel surface compared with BC, attributed to the incorporation of cell affinitive GelMA. By contrast, NIH-3 T3 cells were hardly adhered on the BPEG hydrogel surface. The MTT assay results (Fig. 5F) further demonstrated that the OD value for the BGMA hydrogel group increased after 3 days of culture and was significantly higher than that of the BC group, confirming its positive effect on cell proliferation. However, the BPEG hydrogel group showed a markedly lower OD value that remained nearly unchanged over the same period. The cell affinity of the hydrogels was further investigated through immunofluorescence staining of the cytoskeletal protein F-actin (Fig. 5G). Spindle-shaped NIH-3 T3 cells with massive actin microfilaments were found on the BGMA hydrogel, whereas the expression of actin was strongly suppressed in cells on the BPEG hydrogel. Quantitative analysis indicated greater cell spread area and aspect ratio on the

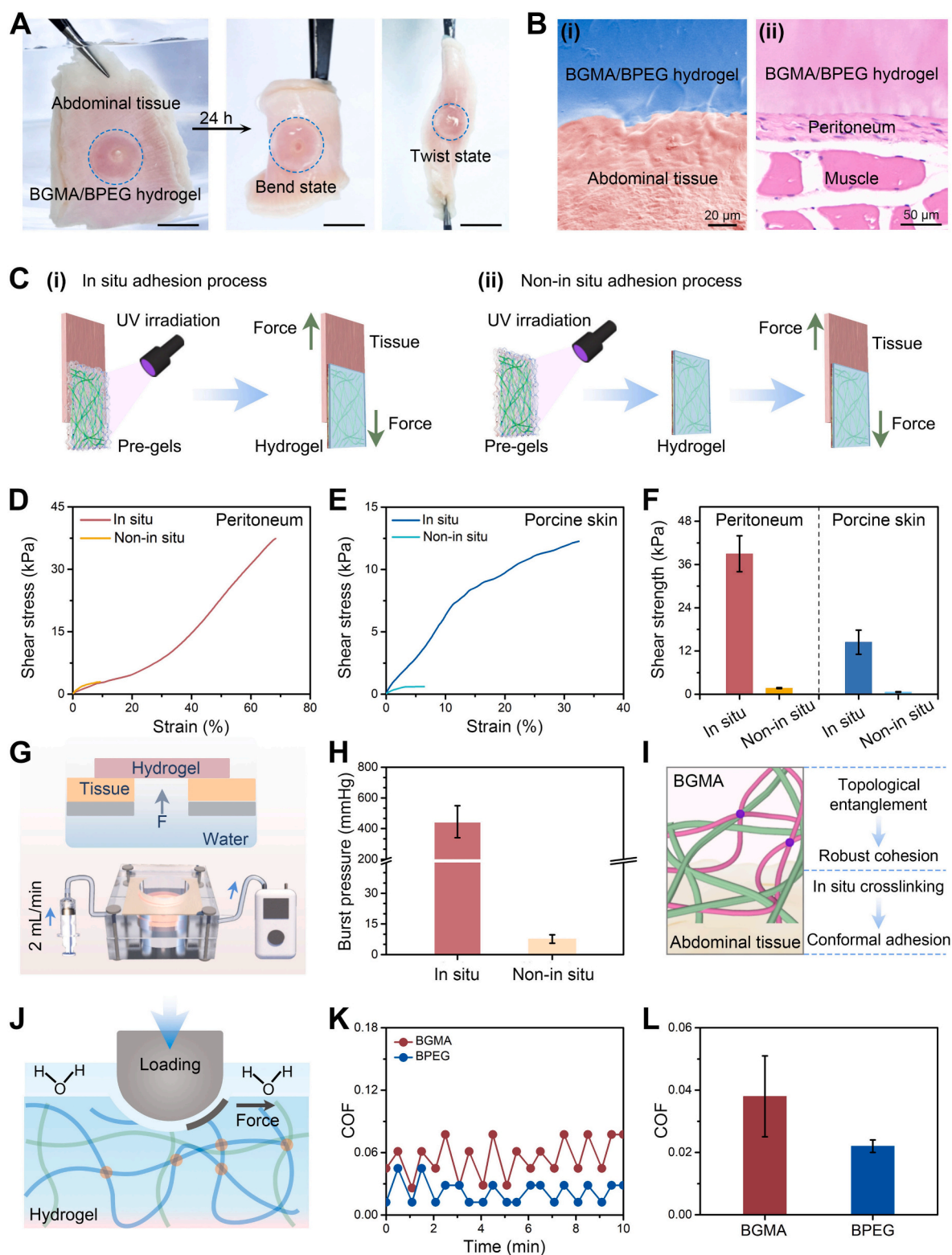


Fig. 4. Asymmetric tissue adhesion of the Janus BGMA/BPEG hydrogel. (A) The BGMA/BPEG hydrogel adhered tightly to the abdominal tissue after 24 h of soaking in PBS, even in a bent or twist state. (B) Adhesion interface between the hydrogel and abdominal tissue observed using (i) SEM and (ii) H&E histological staining. (C) Illustration of the lap-shear tests of (i) the *in situ*-formed hydrogel and (ii) the pre-prepared hydrogel. Shear stress-strain curves of the hydrogels adhered to (D) peritoneum and (E) porcine skin. (F) Shear strength of the hydrogels. (G) Illustration of the burst pressure test. (H) Burst pressure of the hydrogel. (I) Adhesion mechanism of the hydrogel, including *in situ* crosslinking-induced conformal adhesion and topological entanglement-induced robust cohesion. (J) Illustration of the tribological test. (K) COF curves and (L) quantitative analysis of the hydrogels.

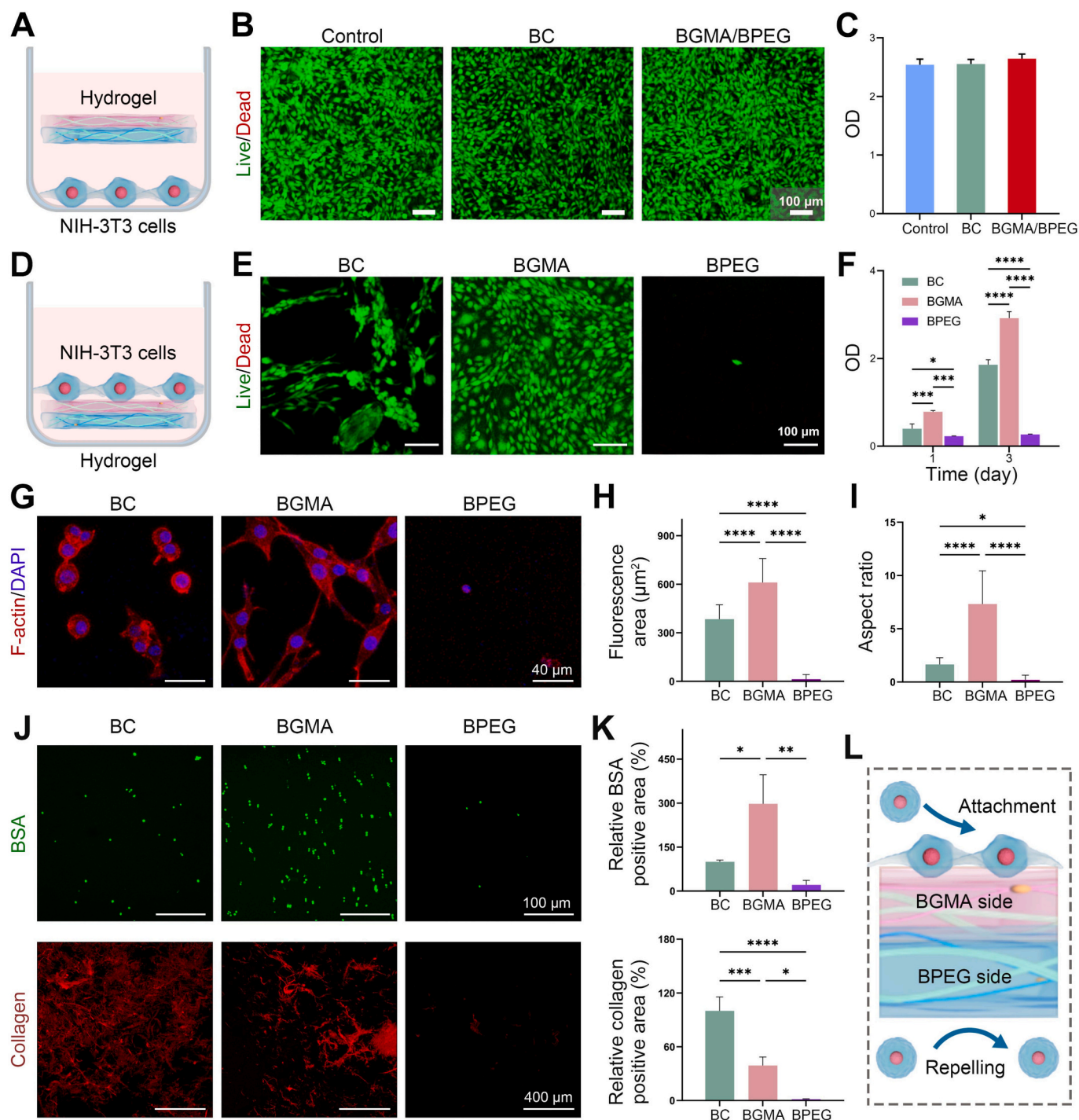


Fig. 5. Asymmetric cell affinity of the Janus BGMA/BPEG hydrogel. (A) Illustration of biocompatibility assessment by seeding NIH-3 T3 fibroblasts on cell plates and subsequently co-culturing them with hydrogels. (B) Live/dead staining of NIH-3 T3 fibroblasts cultured on cell plates. (C) MTT assay for biocompatibility assessment. (D) Illustration of cell affinity evaluation by directly seeding NIH-3 T3 cells on hydrogel surfaces. (E) Live/dead staining of NIH-3 T3 fibroblasts cultured on hydrogels. (F) MTT assay for cell proliferation evaluation. (G) F-actin immunofluorescence staining images of NIH-3 T3 fibroblasts on hydrogel surfaces. (H) Quantitative fluorescence area for actin and (I) aspect ratio of cells cultured on hydrogel surfaces. (J) Fluorescence images of FITC-labeled BSA and collagen adhered to hydrogel surfaces. (K) Quantitative positive area for BSA and collagen. (L) Illustration of the Janus hydrogel with a cell adhesive BGMA side and an anti-adhesive BPEG side. All data are represented as mean $\pm$ SD ($n \geq 3$). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.

BGMA hydrogel compared with the BPEG hydrogel (9-fold and 7-fold higher, respectively), demonstrating that the BGMA hydrogel facilitated cell adhesion but the BPEG hydrogel repelled cells (Fig. 5H, I). To gain deeper mechanistic insight into the cellular adhesion behavior at the protein level, immunofluorescence staining of vinculin, a key focal adhesion protein, was performed. As shown in Fig. S12, NIH-3 T3 cells

cultured on the BGMA hydrogel exhibited significantly stronger fluorescence intensity and a broader distribution area of vinculin proteins, compared to those on the BC and BPEG hydrogels. These protein-level findings provide direct evidence supporting the cell-affinitive properties of the BGMA layer and the effective anti-adhesion characteristics of the BPEG layer. Notably, upon the incorporation of urea into the BGMA

hydrogel network, the cells exhibited a markedly smaller and more rounded morphology, characterized by a significantly diminished spreading area (Fig. S13). This indicates that the disruption of the hydrogen-bonding network by urea compromises the cell affinity of the BGMA hydrogels. In addition to fibroblasts, the BPEG hydrogel also inhibited the attachment of RAW264.7 macrophages (Fig. S14), suggesting its potential to reduce the inflammation-induced adhesions. Given the critical role of proteins in cell adhesion, the adsorption of proteins on both sides of the Janus BGMA/BPEG hydrogel was investigated. Fluorescently labeled bovine serum albumin (BSA) and collagen were selected as representative proteins. As shown in Fig. 5J, K, the BGMA hydrogel adsorbed much more spherical BSA and collagen fibers

than the BPEG hydrogel (14-fold for BSA, 26-fold for collagen), consistent with the phenomenon of cell adhesion. The anti-adhesion properties of the BPEG layer originate from its high hydrophilicity of abundant ether bonds within the PEGDA backbone. This hydrophilicity facilitates to establish a hydrate layer, which prevents direct contact between the BPEG layer and proteins or internal tissues, driving both protein resistance and physical lubrication for anti-adhesion. In short, these results indicate the cell affinitive BGMA side and anti-adhesive BPEG side of the Janus BGMA/BPEG hydrogel (Fig. 5L).

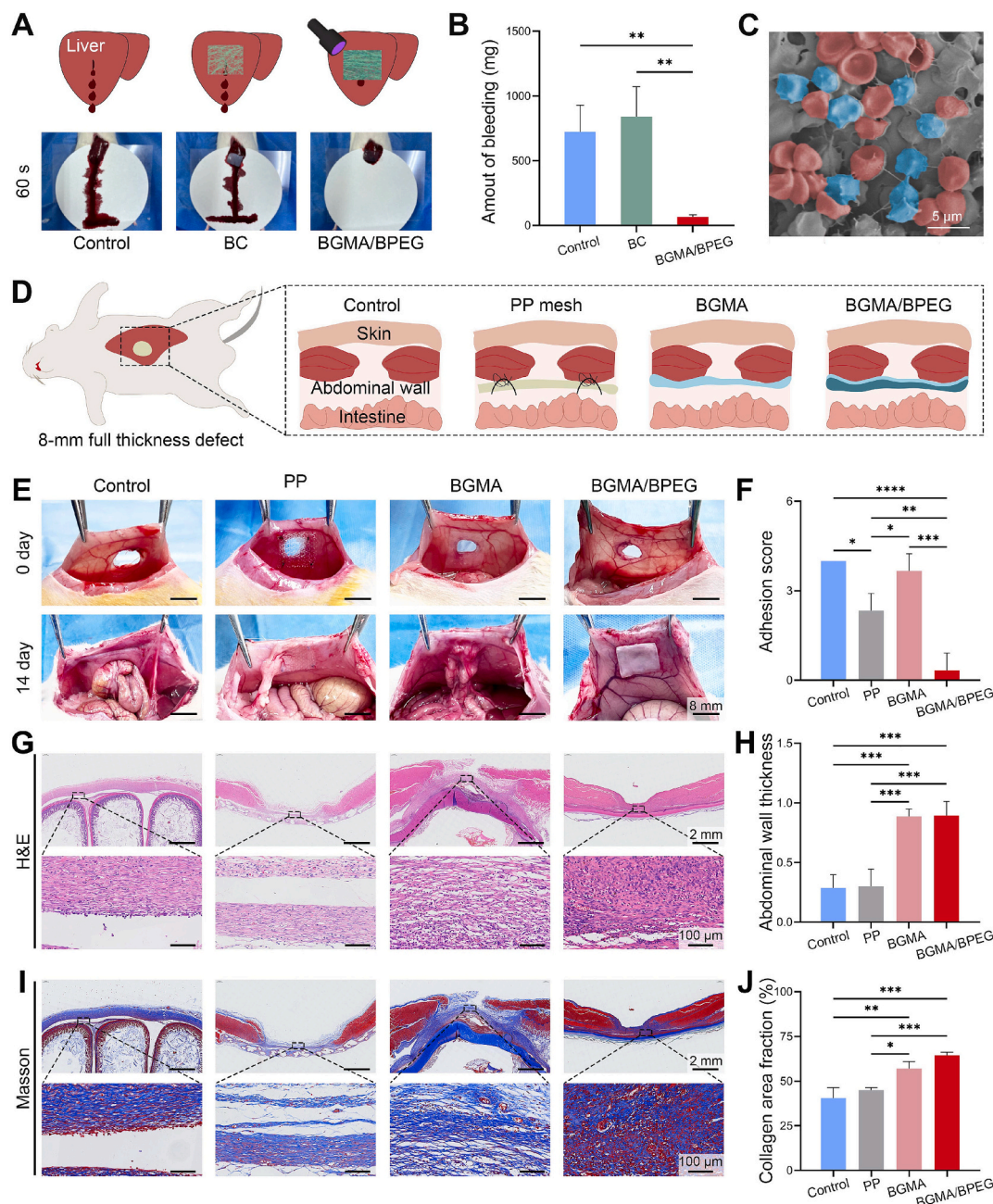


Fig. 6. *In vivo* therapeutic efficacy of the Janus BGMA/BPEG hydrogel for abdominal wall defect treatment. (A) Hemostatic capability of the BGMA/BPEG hydrogel indicated using a liver bleeding model. (B) Quantitative analysis of bleeding amount. (C) SEM images of blood cells adhered on the BGMA side of the Janus hydrogel. Red markers represent red blood cells, and blue markers represent platelets. (D) Establishment of 8-mm full-thickness abdominal defect of rats. (E) Photos and (F) corresponding adhesion scores of abdominal defects following 14 days of treatment. (G) H&E staining and (H) quantitative abdominal wall thickness. (I) Masson staining and (J) quantitative collagen area fraction. All data are represented as mean $\pm$ SD ($n = 3$). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.

2.7. Hemostatic performances of the Janus hydrogels

The Janus BGMA/BPEG hydrogel with one-sided tissue adhesiveness is capable of sealing wounds and arresting bleeding, which contributes to reducing the risk of abdominal wall adhesion caused by hematoma formation. Prior to *in vivo* hemostatic assessment, the hemocompatibility of the BGMA/BPEG hydrogel was evaluated via a hemolysis test. Saline and deionized water was served as the negative and positive controls, respectively. As shown in Fig. S15A, the resultant blood supernatants treated with various amounts of the BGMA/BPEG hydrogel were almost colorless, closely resembling those of the negative control group. Furthermore, the BGMA/BPEG hydrogel group exhibited a hemolysis rate of 0.7%, below the 5% threshold set by international clinical standards [38], suggesting the hemocompatibility of the Janus BGMA/BPEG hydrogel (Fig. S15B).

The local biocompatibility of the Janus BGMA/BPEG hydrogel was evaluated following the direct application of the hydrogel to the liver surface (Fig. S16). H&E staining results revealed that the BGMA group maintained a normal hepatic architecture characterized by regular hepatic plates and well-defined central veins and sinusoids, consistent with observations in the control group. Furthermore, no evident pathological alterations, including tissue necrosis, cellular degeneration, structural disorganization, or substantial inflammatory cell infiltration, were observed. These findings suggest that the hemostatic BGMA hydrogel possesses good *in vivo* biocompatibility and does not trigger local adverse inflammation upon application to the liver surface. The *in vivo* hemostatic efficacy of the hydrogel was next evaluated using a rat liver injury model (Fig. 6A). Photos of the hemostatic process revealed rapid bleeding in untreated livers (control) or BC-treated livers, whereas the BGMA/BPEG hydrogel promptly sealed the wound and dramatically suppressed hemorrhage. Quantitative analysis demonstrated a blood loss of 66 ± 17 mg in the BGMA/BPEG hydrogel group, significantly lower than that of the control (724 ± 205 mg) and BC groups (841 ± 231 mg) (Fig. 6B). Additionally, the BGMA/BPEG hydrogel group exhibited a shorter blood coagulation time (27 ± 7 s), suggesting its superior hemostatic capability (Fig. S17). The hemostatic mechanism of the BGMA/BPEG hydrogel was further evaluated. SEM imaging revealed that the hydrogel efficiently adsorbed red blood cells and promoted platelet activation (Fig. 6C). Moreover, the coagulation index for the BGMA/BPEG hydrogel ($53 \pm 4\%$) was markedly lower than that of the control ($100 \pm 14\%$) and BC groups ($83 \pm 5\%$), indicating a procoagulant effect of the hydrogel (Fig. S18).

2.8. Abdominal anti-adhesion and wound healing performance of the Janus hydrogels

The Janus BGMA/BPEG hydrogel, integrating the aforementioned functionalities, can prevent postoperative adhesion and accelerate wound healing for abdominal defect treatment. To verify this, a rat model of 8-mm full-thickness abdominal wall defect was established. After surgery, the defect area was covered with the *in situ*-formed BGMA or Janus BGMA/BPEG hydrogels. A commercially available polypropylene (PP) mesh sutured directly to the defect site served for comparison. The abdominal defect without treatment served as control (Fig. 6D). Rats were sacrificed 14 days post-surgery for clinical scoring and histological evaluation.

Photos of the abdominal cavity 14 days post-treatment are presented in Fig. 6E. The control group exhibited severe visceral adhesions, including abnormal displacement of gastrointestinal tissue from the abdominal cavity. The BGMA hydrogel acted as a physical barrier that impeded organ leakage but demonstrated limited anti-adhesion efficacy. Although the PP mesh substantially reduced visceral adhesions, persistent adhesions occurred along the lateral aspect of the material, possibly due to incomplete closure resulting from suturing. By contrast, the BGMA/BPEG hydrogel, stably anchored at the wound site throughout the 14-day period of treatment, showed nearly complete absence of

visceral adhesions. Moreover, the average adhesion score for the Janus BGMA/BPEG hydrogel group (0.3 ± 0.6) was significantly lower than those of the control (4 ± 0.0), BGMA hydrogel (3.7 ± 0.6), and PP mesh groups (2.3 ± 0.6) (Fig. 6F). These results indicate the anti-adhesion capability of the Janus BGMA/BPEG hydrogel *in vivo*.

The pro-healing capacity of the Janus BGMA/BPEG hydrogel was analyzed using cross-sectional hematoxylin and eosin (H&E) and Masson histological staining (Fig. 6G, I). In the control group, the abdominal wall defect was occupied by displaced intestinal tissue and exhibited severe inflammatory cell infiltration. Treatment with PP mesh attenuated inflammatory cell presence but generated only a thin fibrous tissue layer at the injury site. Although the BGMA hydrogel group exhibited greater neotissue formation on the hydrogel surface, this neotissue remained separated from the host abdominal wall tissues, possibly due to the detachment of hydrogel caused by visceral adhesions. By contrast, the Janus BGMA/BPEG hydrogel demonstrated conformal attachment to abdominal tissues, promoting integration between neotissue and host tissue while accelerating collagen deposition. Notably, the magnified H&E staining image demonstrates the seamless connection between the two layers of the Janus BGMA/BPEG hydrogel (Fig. S19). Quantitative analysis (Fig. 6H, J) indicated that BGMA/BPEG hydrogel treatment significantly promoted the neotissue thickness and collagen deposition compared with that of the control (3.1-fold higher and 1.6-fold higher, respectively) and PP mesh group (3.0-fold higher and 1.4-fold higher, respectively), demonstrating rapid and high-quality healing. The effect of the hydrogels on abdominal wall healing was further evaluated from a planar perspective (Fig. S20). The control group displayed severe intestinal leakage; consequently, the wound expanded beyond its original size, making the initial boundaries unidentifiable. The PP mesh failed to significantly promote wound healing at 14 days post-implantation, with a wound healing ratio of $59.2\% \pm 6.4\%$. By contrast, the wound healing ratio for the BGMA and BGMA/BPEG groups was increased to $91.8\% \pm 5.9\%$ and $95.0\% \pm 6.9\%$, respectively. This demonstrates the positive effects of the BGMA/BPEG hydrogel on wound healing, which can be attributed to the cell-affinitive GelMA component within the hydrogel matrix.

The *in vivo* inflammation and immune response were evaluated using immunofluorescence staining for iNOS and CD206 (Fig. S21), markers representing pro-inflammatory M1 and pro-healing M2 macrophages, respectively. The control group exhibited strong iNOS and minimal CD206 fluorescence signals, indicating a severely pro-inflammatory microenvironment. Treatment with the PP mesh had a negligible effect on the expression of these two markers. By contrast, both the BGMA and the Janus BGMA/BPEG hydrogels significantly reduced iNOS expression and substantially increased CD206 signals. Quantitative analysis revealed that the Janus hydrogel group displayed the lowest iNOS levels and the highest CD206 levels, demonstrating the transition from a pro-inflammatory microenvironment to a pro-healing state. This can be attributed to the synergistic effects of seamless sealing, the prevention of fibrous adhesion, and the inherent immunomodulatory capacity of the GelMA component within the hydrogel [39].

The *in vivo* biocompatibility of the BGMA/BPEG hydrogel was next assessed. Histological examination via H&E staining revealed no apparent signs of toxicity in the major organs, including heart, liver, spleen, lung, and kidney (Fig. S22). Hematology analysis indicated the levels of the WBC, RBC, HGB, HCT, MCV, and MCH of the BGMA/BPEG hydrogel group were comparable to those of the control group (Fig. S23). Additionally, the treatment with the BGMA/BPEG hydrogel did not cause significant abnormalities in blood biochemical profiles (AST, ALT, CREA-S, UREA) (Fig. S24). Collectively, these results demonstrated the good biocompatibility of the BGMA/BPEG hydrogel.

The superior therapeutic efficacy of the Janus BGMA/BPEG hydrogel can be assigned to the following four aspects. First, the adhesive BGMA side allows conformal adherence to the abdominal wall defect, seamlessly closing the wound and minimizing the risk of leakage. Second, the BGMA side is cell affinitive, promoting fibroblast adhesion and

proliferation to accelerate wound healing. Third, the BPEG side repels cellular and protein attachment, reducing the formation of abdominal adhesions. Fourth, the BGMA/BPEG hydrogel with topological entanglement structure exhibits mechanical robustness and minimal swelling, facilitating the maintenance of its structural integrity and protecting interfacial adhesion within the biofluid-rich and mechanically dynamic abdominal environment during treatment.

3. Discussion

Seamless closure using adhesive hydrogel patches is essential for preventing hernia formation and reducing infection risks during abdominal wall defect repair. However, these hydrogels inherently suffer from an adhesion-cohesion conflict. For example, a higher crosslinking density within the hydrogel patch limits polymer chain mobility, leading to insufficient interfacial binding with abdominal tissues [40,41]. Conversely, decreasing the crosslinking density improves interfacial binding but sacrifices swelling resistance [42], ultimately resulting in adhesion failure. Recent efforts have focused on developing *in situ*-formed adhesive hydrogels by injecting precursor solutions or spraying freeze-dried hydrogel powders onto the tissues [43,44]. While these *in situ*-formed hydrogels exhibit favorable interfacial binding due to the high mobility of polymer chains prior to crosslinking, their precursors or powders are prone to leaking from the defect site. This is due to the absence of mechanical support, particularly in typically large abdominal wall defects. Furthermore, most existing adhesive hydrogels are relatively brittle and susceptible to structural rupture upon abdominal tissue deformation.

In the current study, the adhesive BGMA hydrogel is prepared by physically immobilizing GelMA chains within a BC nanofibril framework via topological entanglement, followed by photo-induced *in situ* crosslinking. This strategy effectively prevents the outflow of GelMA chains from the pre-gel, ensuring stable localization at the abdominal defect site. Crucially, the pre-gel maintains sufficient GelMA chain mobility to establish a tight initial attachment with the abdominal tissue. This attachment is subsequently locked in place via photo-induced crosslinking, creating strong interfacial adhesion for the seamless closure of the defect. Additionally, the topological entanglement structure promotes energy dissipation, remarkably enhancing the toughness of the BGMA hydrogel. Together with covalent crosslinking, this physical reinforcement allows the hydrogel to maintain its structural integrity within the mechanically dynamic abdominal environment. Furthermore, the combination of topological entanglement and covalent crosslinking restricts polymer chain relaxation, contributing to a low swelling equilibrium. Overall, the proposed topological entanglement-reinforced hydrogel demonstrates enhanced adhesion strength, superior mechanical toughness, and a low swelling ratio, making it highly suitable for the seamless closure of abdominal defects in biofluid-rich and mechanically dynamic environments.

4. Conclusions

A topological entanglement-reinforced Janus BGMA/BPEG hydrogel with asymmetric adhesion, mechanical robustness, and anti-swelling properties was constructed for abdominal wall defect repair. This topological entanglement physically immobilized photocrosslinkable GelMA or PEGDA polymer chains within the BC framework, enabling *in situ* crosslinking at the wound site to establish strong interfacial binding to abdominal tissues with 39.0 ± 5.0 kPa shear strength. Serving as an efficient pathway for load transfer and energy dissipation, the topological entanglement structure endowed the Janus hydrogel with 1.55 ± 0.13 MPa tensile strength and 0.36 ± 0.05 MJ/m³ toughness. Furthermore, topological constraints restricted chain relaxation, resulting in a low equilibrium swelling ratio of $0.22\% \pm 0.08\%$. This topological entanglement-mediated reinforcement mechanism overcomes the limitation of adhesive hydrogels experiencing adhesion-cohesion conflict,

facilitating to maintaining structural integrity and interfacial adhesion within the biofluid-rich and mechanically dynamic abdominal environment. *In vivo* experiments demonstrated that the Janus hydrogel seamlessly closed the abdominal defect, accelerated wound healing, and prevented visceral adhesions after 14 days. This superior therapeutic efficacy of the Janus BGMA/BPEG hydrogel surpassed that of the commercially available PP patch, highlighting its potential as a promising preclinical abdominal patch. Despite these promising outcomes, several challenges remain and require further investigation prior to clinical translation. Specifically, future studies must establish a precise relationship between the hydrogel's *in vivo* degradation behavior and its functional performance, such as long-term adhesion retention and mechanical stability. Furthermore, while the subcutaneous implantation model provided valuable initial degradation data, it does not fully replicate the complex biomechanical and biofluid-rich environment of actual abdominal wall defects. Finally, the long-term behavior, degradation kinetics, and safety profile of the hydrogel beyond the reported 14-day time frame remain to be systematically evaluated. Overall, this work provides a new paradigm for designing suture-free, anti-adhesion hydrogel patches for soft tissue repair.

5. Experimental section

5.1. Materials

Acetobacter xylinum was kindly provided by Tianjin University of Science and Technology (Tianjin, China). D-(+)-glucose anhydrous (BioFroxx), yeast extract and tryptone (Oxoid) were used for bacterial cellulose culture. Type A gelatin from porcine skin (Cat. No. G1890), poly(ethylene glycol) diacrylate (PEGDA, Mn = 700 Da), methacrylic anhydride, and rhodamine B were purchased from Sigma-Aldrich. Lithium phenyl(2,4,6-trimethylbenzoyl) phosphinate (LAP) was sourced from Engineering For Life (Jiangsu, China). Anhydrous disodium hydrogen phosphate and glacial acetic acid were obtained from Kelong Chemical (Chengdu, China). Fluorescein isothiocyanate (FITC) was acquired from Aladdin (Shanghai, China), and FITC-conjugated bovine serum albumin (FITC-BSA) was purchased from Beijing Boerxi Technology Co., Ltd. (Beijing, China).

5.2. Synthesis of the bacterial cellulose

Bacterial cellulose (BC) was produced by *Acetobacter xylinum*. An activation culture was first incubated statically at 30 °C for 3 days in a sterile medium containing 2.5% (w/v) D-(+)-glucose, 0.75% yeast extract, 1.0% tryptone, 1.0% Na₂HPO₄, and 1.0% (v/v) acetic acid. Subsequently, 8 mL of the activated culture was transferred to a 6 cm Petri dish and cultured for 4 days to form BC pellicles. Harvested pellicles were purified by immersion in 1% (w/v) NaOH for 12 h, followed by 10 alternating washing cycles (12 h each) with deionized water and 75% (v/v) ethanol. The purified BC films were stored in 75% (v/v) ethanol.

5.3. Synthesis of the methacrylated gelatin (GelMA)

Methacrylated gelatin (GelMA) was prepared as follows. Gelatin (5 g) was dissolved in 50 mL of PBS (pH 7.4) at 50 °C. Methacrylic anhydride (5 mL) was added dropwise, and the mixture was reacted for 2 h at 50 °C. The reaction was quenched by dilution with 200 mL of PBS. The product was purified by dialysis (MWCO 3500 Da) against deionized water at 50 °C for 7 days, followed by lyophilization to obtain GelMA.

5.4. Synthesis of the BC-GelMA hydrogel (BGMA) and BC-GelMA pre-gel (BGMA pre-gel)

40% (w/v) GelMA precursor solution was prepared by dissolving GelMA in an aqueous solution of 0.05% (w/v) LAP photoinitiator at

60 °C. To prepare the BGMA pre-gel, pre-cut (10 × 20 mm) BC pellicles, blotted to remove surface water, were soaked in this precursor solution at 60 °C for 24 h. After carefully removing the excess liquid from the surface, the infiltrated composite sheets were obtained. To form the final BGMA hydrogel, these pre-gels were immediately crosslinked via UV irradiation (395 nm, 20 s).

5.5. Synthesis of BC-PEGDA hydrogel (BPEG) and BC-PEGDA pre-gel (BPEG pre-gel)

To prepare the BPEG pre-gel, a precursor solution was formulated containing 16.7% (v/v) PEGDA and 0.05% (w/v) LAP in deionized water. Pre-cut (10 × 20 mm) bacterial cellulose (BC) pellicles, with excess surface water removed, were then immersed in this solution and soaked for 24 h at room temperature. To obtain the BPEG hydrogel, the resulting pre-gel were removed from the solution, gently blotted to remove surface precursor, and immediately irradiated with UV light (395 nm, 20 s), which initiated photo crosslinking.

5.6. Synthesis of Janus BGMA/BPEG hydrogel

First, BGMA pre-gel and BPEG pre-gel were prepared as described above. Then, the BPEG pre-gel was layered onto the BGMA pre-gel. Last, the BGMA/BPEG Janus hydrogel were obtained by photo crosslinking under 395 nm irradiation.

5.7. Characterizations

Details are available in the Supporting Information.

CRediT authorship contribution statement

Ting Zhou: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Liwei Yan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Wei Zhang:** Methodology, Data curation, Conceptualization. **Nini Xin:** Methodology, Data curation. **Jiamei Xiao:** Methodology, Investigation. **Jie Ding:** Methodology, Investigation. **Chengheng Wu:** Methodology, Investigation. **Dan Wei:** Investigation, Funding acquisition. **Jing Sun:** Writing – review & editing, Writing – original draft, Visualization, Funding acquisition, Conceptualization. **Hongsong Fan:** Writing – review & editing, Writing – original draft, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Data availability

Data will be made available on request.

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